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Quadriceps tendon autograft is not inferior to bone-patellar tendon-bone or hamstring autograft in anterior cruciate ligament reconstruction in terms of subjective patient-reported outcomes: Two-year results of a multicenter randomised controlled study

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Abstract

Purpose: To evaluate the effectiveness of quadriceps tendon (QT) compared to bone-patellar tendon-bone (BPTB) and hamstring tendon (HT) autografts in anterior cruciate ligament reconstruction (ACLR) regarding patient-reported outcomes 2 years postoperatively.

Methods: Patients undergoing primary isolated ACLR were stratified and randomised to receive a QT, BPTB or HT autograft. The primary outcome was the International Knee Documentation Committee (IKDC) subjective score analysed to assess non-inferiority at 2-year follow-up. Secondary outcomes included Knee Injury and Osteoarthritis Outcome Score, Anterior Cruciate Ligament Quality of Life, Lysholm, Tegner, pain and satisfaction scores, IKDC physical examination score, limb symmetry indices, radiographic osteoarthritis and adverse events including failure.

Results: One hundred and thirty-six patients were analysed at 2-year follow-up (50 QT, 42 BPTB, 44 HT); 68%, 62% and 64% were male, with mean ages of 26.4, 25.9 and 25.8 years, respectively. The lower limit of the 95% confidence interval (CI) for the IKDC subjective score for the QT group (95% CI 85.4–91.5) fell within the prespecified noninferiority margin (–10). No statistical between-group differences were found for mean IKDC subjective scores (QT, 88.4 ± 8.6; BPTB, 85.5 ± 12.9; HT, 89.2 ± 12.1). All groups showed significant improvement in patient-reported outcomes over time ($p < 0.001$), with no significant between-group differences. The HT

Abbreviations: ACL, anterior cruciate ligament; ACLR, anterior cruciate ligament reconstruction; ACL-QOL, anterior cruciate ligament quality of life; ADL, activities of daily living; ALCR, anterolateral corner reconstruction; ANOVA, analysis of variance; AP, anteroposterior; BPTB, bone–patellar tendon–bone; CI, confidence interval; CONSORT, consolidated standards of reporting trials; HT, Hamstring tendon; IC, informed consent; IKDC, International Knee Documentation Committee; KL, Kellgren–Lawrence; KOOS, knee injury and osteoarthritis outcome score; LSI, limb symmetry index; LTFU, lost to follow-up; MCID, minimal clinically important difference; METC, Medical Ethical Committee; MRI, magnetic resonance imaging; NRS, numerical rating scale; n.s., not significant; PASS, patient acceptable symptom state; PROMs, patient-reported outcome measures; QoL, quality of life; QT, quadriceps tendon; RCT, randomised controlled trial; SD, standard deviation; SPSS, Statistical Package for the Social Sciences; VAS, visual analog scale.

group demonstrated superior quadriceps strength and triple hop performance at 6 months (all $p \le 0.035$), which resolved by 1 year. Four clinical failures occurred in the QT group (8.0%), five in the BPTB group (11.9%) and two in the HT group (4.6%) (n.s.).
Conclusion: QT autograft is non-inferior to BPTB or HT autograft in isolated ACLR in terms of patient-reported outcomes as measured with the IKDC subjective score two years postoperatively. QT autograft represents a viable option for ACLR.
Trial Registration: ClinicalTrials.gov identifier: NCT03073083.

Level of Evidence: Level I.

KEYWORDS
ACL reconstruction, anterior cruciate ligament, bone–patellar tendon–bone, hamstring tendon, quadriceps tendon

INTRODUCTION

Graft choice in anterior cruciate ligament reconstruction (ACLR) remains a subject of ongoing debate. Historically, the bone-patellar tendon-bone (BPTB) autograft is considered the gold standard, while the hamstring tendon (HT) autograft has become the most commonly used option globally [2, 43]. In recent years, the quadriceps tendon (QT) autograft has gained popularity due to its favourable biomechanical properties and potentially reduced donor site morbidity [2].

Numerous recent randomised controlled trials (RCTs), registry studies and meta-analyses have demonstrated comparable patient-reported outcomes between ACLR with BPTB or HT autografts [4, 7, 32, 36]. Although several studies suggest comparable outcomes for ACLR with QT autografts relative to BPTB or HT autografts, these findings stem largely from indirect comparisons across heterogeneous study populations [1, 7, 21, 22, 24, 34]. BPTB grafts may have lower failure rates than HT in young, active patients but are associated with higher donor-site morbidity, whereas QT grafts may reduce donor-side morbidity with evidence of higher failure rates compared with BPTB [23, 41]. A recent review highlighted that existing RCTs (11 studies) comparing ACLR with QT autograft to other autografts are statistically fragile [10], and to date, no prospective randomised study has directly compared all three autografts within a single trial [1, 26].

Graft choice, beside other surgical strategies as treatment of associated pathology and the more recently introduced anterolateral corner reconstruction (ALCR), can influence outcome [40, 44, 45]. Direct comparisons within a single study are essential to guide evidence-based, individualised graft selection in clinical practice [33]. Therefore, the purpose of this study was to directly compare patient-reported, clinical and radiological outcomes following isolated ACLR with QT, BPTB and HT autografts. The aim was to examine the following null hypothesis: ACLR with a QT autograft would be non-inferior to ACLR with a BPTB or HT autograft in terms of patient-reported outcomes 2 years postoperatively.

METHODS

Design

A prospective multicenter RCT was conducted at the Centre for Orthopaedic Surgery OCON, Hengelo, the Netherlands; Martini Hospital, Groningen, the Netherlands; and Gelderse Vallei Hospital, Ede, the Netherlands. The study was approved by the Medical Ethical Committee (METC) (NL52749.044.16) and the institutional review boards of all three collaborating hospitals, and is reported according to the CONSORT statement [37].

The trial was originally designed with graft failure at 2 years postoperatively as the primary outcome with a planned enrolment (noninferiority margin, 12%; $\alpha = 0.0125$; power, 80%) of 146 patients per group (total $n = 439$). Enrolment was prematurely terminated after 136 participants because of delayed recruitment, driven by patients' preference to participate in graft selection, the corona pandemic and the adoption of additional anterolateral corner reconstructions during the course of the study, while the study protocol was limited to primary ACLR without concomitant ALCR. Consequently, the study was underpowered to detect differences in graft failure. Amending the protocol to allow ALCR would have required a fundamentally different study design and sample size to adequately assess graft failure, precluding inclusion of patients already enrolled. The achieved sample size, however, provided >90% a-priori power to detect a clinically meaningful difference in the International Knee Documentation Committee (IKDC) 2000 subjective score at 2 years, which was therefore analysed as the primary outcome. As the first randomised controlled trial including all three graft types, this study offers valuable clinical insights despite being underpowered for graft failure. The present report therefore presents the IKDC subjective score as the primary and graft failure as a secondary exploratory outcome.

Patients

Patients aged ≥18 years were enrolled by their treating orthopaedic surgeon at the outpatient department (CONSORT flowchart Figure 1). The following inclusion criteria were used for selection: (1) primary ACL rupture confirmed through patient history, physical examination and magnetic resonance imaging (MRI); (2) an indication for ACLR set within 6 months post-injury (as was the norm at the time of conception of the study protocol); [38] and (3) a Tegner Activity Scale score of ≥5 [42]. Exclusion criteria were: (1) concomitant ligamentous injury; (2) meniscal lesions needing surgical repair and (3) full-thickness cartilage lesions, as these injuries require adjustments of the postoperative rehabilitation regime; (4) osteoarthritis grade ≥2 according to Kellgren-Lawrence Score; [20] (5) ipsilateral leg malalignment (varus/valgus ≥ 5 degrees; posterior tibial slope ≥ 12 degrees); (6) prior ipsi- or contralateral knee

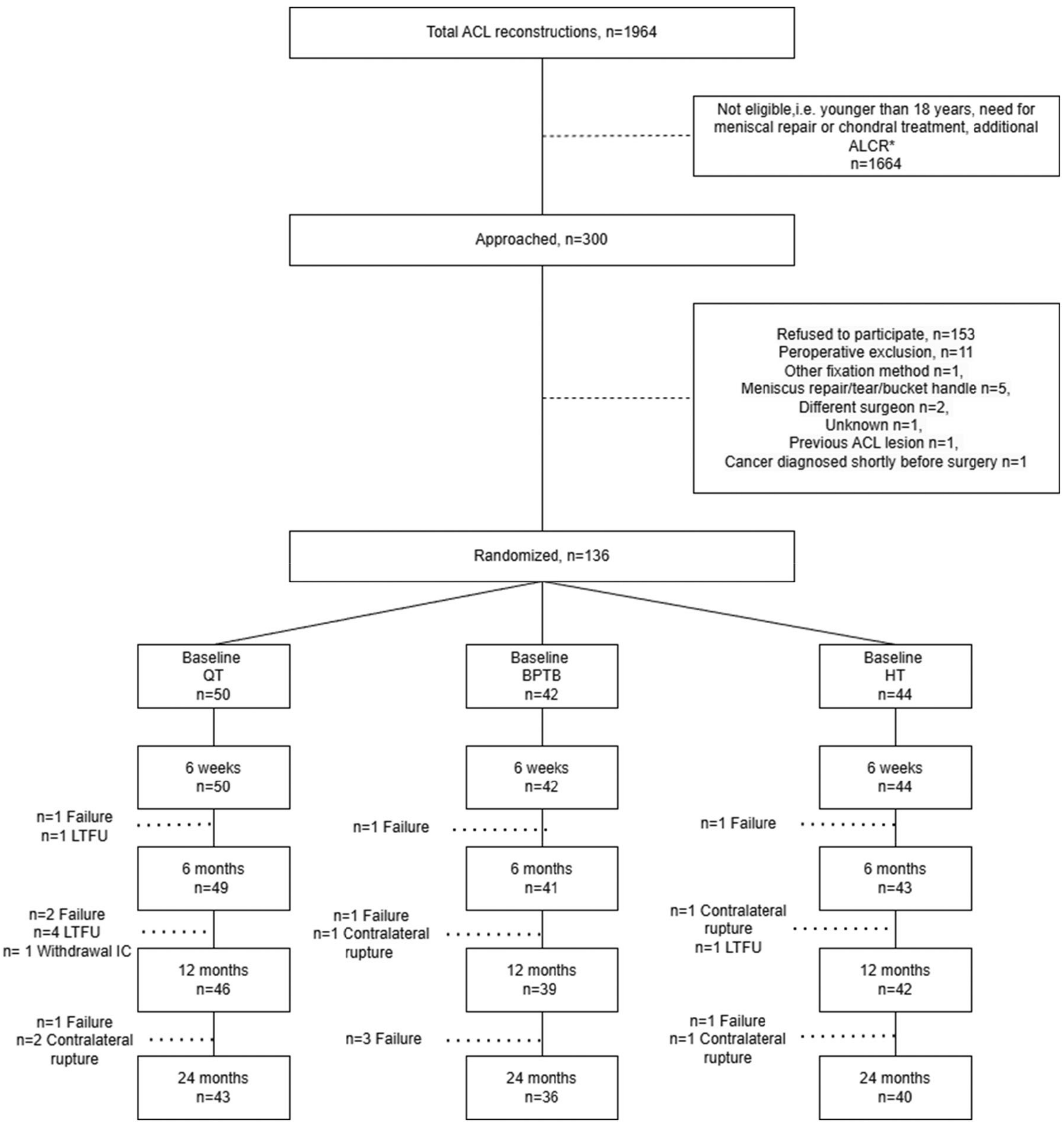

FIGURE 1 CONSORT flowchart inclusion and randomisation of subjects. *After 2021, inclusion declined following the standardisation of ACLR in high-risk patients. ACL, anterior cruciate ligament; ALCR, anterolateral corner reconstruction; IC, informed consent; LTFU, lost to follow-up.

surgery; (7) surgery affecting graft harvest sites; (8) hypersensitivity to implant materials; (9) neuromuscular disorders or vascular disorders affecting rehabilitation; (10) abnormal scar formation; (11) active infection; (12) pregnancy; (13) osteoporosis or (14) chronic use of prednisone or cytostatics.

Randomisation

Following written informed consent, patients were stratified by age (18–25 years of age versus >25 years of age), preinjury Tegner Activity Score (intermediate: 5–7; high: 8–10), and per operating orthopaedic surgeon to balance reinjury risk and graft allocation. Block randomisation using sealed envelopes (computer-generated, variable block sizes $n = 2$, 4, 6) was preoperatively performed by an independent researcher to allocate patients to undergo ACLR with either a QT (QT-group), BPTB (BPTB-group) or HT (HT-group) autograft. Due to practical constraints, neither patients nor outcome assessors were blinded to the treatment allocation.

Intervention

As per protocol, all surgical procedures were performed within 6 weeks after the indication was set. After performing routine arthroscopy to assess all compartments for concomitant injuries and confirming the absence of exclusion criteria, the appropriate graft was harvested. For all surgical techniques, the femoral tunnel was placed in the ACL's anteromedial bundle attachment site. The tibial tunnel was placed centrally in the ACL attachment site, using the optical centre as well as the posterior side of the anterior horn of the lateral meniscus as references.

ACLR with ipsilateral HT autograft

ACLR with an ipsilateral semitendinosus HT autograft was performed with an all-inside technique using variable loop length cortical suspensory fixation (TightRope, Arthrex) [25]. The graft was harvested with a mini-incision technique at the posterior side of the knee and quadrupled [31]. The length of the graft was adjusted so, that the graft would fill a 15 mm femoral socket and would fill at least 20 mm of the tibial socket. Independent all-inside tibial and femoral sockets were prepared with a retrograde drill (FlipCutter) [25]. The graft was tensioned with the knee at 0° of knee flexion to prevent overconstraining, and manual posterior reduction of the tibia relative to the femur. Adequate graft tension was confirmed under arthroscopic view, and the graft was re-tensioned if deemed appropriate.

ACLR with ipsilateral QT autograft

Technically, ACLR with ipsilateral QT autograft was performed identically to ACLR with ipsilateral semitendinosus autograft. The graft was harvested with an anterior incision just proximal to the proximal pole of the patella, using a cutting blade to provide a 10 mm wide strip. Any capsular defect was closed, and with full-thickness graft harvests, the harvest defect in the QT was closed.

ACLR with ipsilateral BPTB autograft

ACLR with an ipsilateral BPTB autograft was performed using the accessory anteromedial portal technique. The graft of 10 mm was harvested from the middle-third of the patellar tendon with a bone block of approximately 20 mm in length and 10 mm × 10 mm in depth and width on both sides of the graft via an anterior 2-incision technique. Independent tibial and femoral sockets were prepared with a low-profile drill (Arthrex). The graft was positioned in the femoral tunnel with the tendon parallel to the direct insertion of the ACL fibres [39] with the bone block facing antero-inferior, and fixed with a titanium interference screw (Smith & Nephew). The graft was fixed in the tibial tunnel with a BioComposite interference screw (Arthrex) with the knee in 20° of flexion, maximal manual traction on the distal end of the graft, and manual posterior reduction of the tibia relative to the femur.

All procedures were performed by five orthopaedic surgeons specialised in ligamentous knee surgery, with 6–22 years of experience, an annual volume of 100–350 ACL reconstructions and familiarity with all three ACLR graft types.

Postoperative rehabilitation

All patients followed a structured, criteria-based rehabilitation protocol supervised by their own physical therapist [46]. The protocol was based on national guidelines for ACL rehabilitation, emphasising strength restoration, neuromuscular control and functional criteria for return to sports. Adherence to the protocol was checked at every follow-up moment, including written feedback of the treating physiotherapist.

Baseline characteristics

Patient baseline and perioperative characteristics comprised sex, age, injured side, body mass index, smoking status, time from injury to surgery, presence of concomitant cartilage and meniscal injuries and operating time.

Primary outcome measure: IKDC subjective at 2-year follow-up

The primary outcome measure was the International Knee Documentation Committee 2000 (IKDC) subjective score 2 years postoperatively to assess symptoms and functional limitations of the knee [18].

Secondary outcome measures

Patient-reported outcome measures (PROMs) and physical examinations were assessed at baseline and at 6 weeks, and 6, 12 and 24 months postoperatively. PROMs included the IKDC 2000 subjective score (patient-acceptable symptom state [PASS] threshold 75.9); [18, 27] the Knee injury and Osteoarthritis Outcome Score (KOOS) to evaluate pain, symptoms, activities of daily living, sport and recreation function and knee-related quality of life (PASS threshold 88.9, 57.1, 100, 75 and 62.5 respectively); [15, 27] the anterior cruciate ligament quality of life (ACL-QOL) score to evaluate the impact of ACL injury on the quality of life; [9] the Tegner activity score to evaluate physical activity level; [42] the Lysholm score to evaluate knee specific symptoms (PASS threshold 85); [5, 29] a visual analog scale (VAS) to evaluate the level of anterior knee pain at rest, during exercise and during kneeling and to assess instability; a numeric rating score (NRS) to evaluate overall satisfaction with surgical outcome [11]. Physical examination included the IKDC physical examination score and instrumented Lachman testing (Rolimeter, Aircast Europe) [13].

The leg symmetry index (LSI) for isokinetic quadriceps and hamstrings strength (Isoforce dynamometer; TUR) (peak torque at 60, 180 and 300 deg/s) and for hop tests (single legged hop and hold, side hop and triple hop for distance) were evaluated at 6, 12 and 24 months postoperatively [16, 28]. LSI for strength was also evaluated pre-operatively at baseline. All clinimetric assessments were performed by experienced sports-physiotherapists in the outpatient clinic of the orthopaedic department. The LSI was calculated as (injured side/uninjured side) × 100%. All tests were administered by experienced sports physical therapists following standardised protocols. All equipment was calibrated prior to each measurement session.

Osteoarthritis was classified according to the Kellgren–Lawrence (KL) score on the anteroposterior weightbearing knee radiographs at baseline and at 12 and 24 months postoperatively.

Clinical failure, repeat surgery and other complications or adverse events were documented and retrieved from patient records. Clinical failure was defined as subjective instability, pathological graft laxity (instrumented Lachman >1+ and/or instrumented anterior drawer >1+, and/or pivot shift >0 compared to the contralateral side) and/or graft discontinuity detected on MRI or during arthroscopy.

Statistical analysis

The primary analysis was a non-inferiority comparison of the International Knee Documentation Committee (IKDC) subjective score at 2 years follow-up between ACLR with QT and both BPTB and HT autografts. A non-inferiority margin (Δ) of –10 points was chosen, based on previously reported minimal clinically important differences (MCID) ranging from 8.8 to 15.6 [6]. Assuming a standard deviation of 10 points, a sample size of 23 patients per group would provide 90% power with a one-sided alpha of 0.008.

Descriptive results were presented as mean (±SD) or frequencies (%). Group comparisons were performed using analysis of variance (ANOVA) or Kruskal–Wallis tests for continuous variables and chi-square or Fisher's exact tests for categorical variables. Post-hoc pairwise comparisons were conducted using Tukey's HSD (when homogeneity of variances was met) or Games–Howell tests (when homogeneity of variances was violated). For categorical variables, pairwise chi-square tests with Bonferroni correction were applied where necessary.

Non-inferiority was assessed using per-protocol analysis. ACLR with a QT autograft was considered non-inferior if the lower boundary of the two-sided 95% confidence interval (CI) of the IKDC subjective score was within 10 points of the mean score of the comparator group. Secondary outcomes were analysed using mixed-effects models to evaluate group, time and group-by-time interaction effects. A random intercept for each participant was included to account for repeated measurements. A p-value < 0.05 was considered statistically significant. Statistical analyses were performed using SPSS version 28.0 (SPSS Inc.).

RESULTS

Between March 2017 to March 2022, 136 patients were randomised to receive ACLR with a QT ($n = 50$, BPTB ($n = 42$) or HT ($n = 44$) autograft (Figure 1). During the 2-year follow-up, five patients in the QT group were lost to follow-up despite multiple attempts to contact them, and one patient withdrew his consent after enrolment. In the HT group one patient was lost to follow-up. The COVID-19 pandemic resulted in incomplete collection of clinical outcome data across time points between 2020 and 2022.

Baseline characteristics

Baseline characteristics are presented in Table 1. No differences were found in baseline characteristics between groups, except for a significant shorter operating time for the HT-group compared to both the QT-group (mean difference 15.4 min) and BPTB-group (mean difference 14.6 min; $p < 0.001$) and a significant higher KOOS ADL

TABLE 1 Baseline characteristics and preoperative findings.

	QT (*n* = 50)	BPTB (*n* = 42)	HT (*n* = 44)	*p*-value
Sex				0.856
Men	34 (68%)	26 (62%)	28 (64%)	
Women	16 (32%)	16 (38%)	16 (36%)	
Age (years)	26.4 ± 6.5	25.9 ± 6.9	25.8 ± 6.2	0.920
Age stratification				
18–25 years	27 (54.0%)	24 (57.1%)	26 (59.1%)	0.881
>25 years	23 (46.0%)	18 (42.9%)	18 (40.9%)	
Injured side				0.935
Left	22 (44%)	20 (48%)	21 (48%)	
Right	28 (56%)	22 (52%)	23 (52%)	
Body mass index	24.2 ± 3.0	24.1 ± 2.8	24.3 ± 2.8	0.963
Smoking				0.192
Yes	9 (18%)	8 (19%)	16 (36%)	
No, never	41 (82%)	34 (81%)	28 (64%)	
IKDC subjective	60.1 ± 13.7	61.2 ± 14.1	56.9 ± 13.1	0.223
KOOS				
Other symptoms	68.2 ± 18.2	70.7 ± 23.5	67.1 ± 18.3	0.697
Pain	73.4 ± 15.6	78.9 ± 12.6	74.0 ± 16.6	0.179
ADL	85.4 ± 14.3	90.3 ± 9.7	82.7 ± 14.6	0.030
Sports and recreation	44.4 ± 23.0	46.1 ± 26.2	38.1 ± 27.0	0.313
Knee-related QoL	47.3 ± 14.9	50.4 ± 18.1	46.7 ± 17.6	0.541
ACL-QOL, total	45.4 ± 14.7	50.5 ± 17.4	49.1 ± 14.9	0.271
Lysholm	69.3 ± 14.7	71.3 ± 14.5	71.4 ± 14.7	0.725
Tegner	8.0 ± 1.2	7.9 ± 1.3	7.7 ± 1.1	0.582
Tegner stratification				0.610
Intermediate (5–7)	23 (46%)	20 (48%)	24 (56%)	
High (8–10)	27 (54%)	22 (52%)	19 (44%)	
IKDC physical examination score				0.612
A	0 (0%)	1 (2%)	0 (0%)	
B	4 (8%)	2 (5%)	4 (9%)	
C	42 (84%)	32 (76%)	34 (77%)	
D	4 (8%)	7 (17%)	6 (14%)	
Lachman delta	3.4 ± 2.03	3.66 ± 1.76	4.06 ± 1.86	0.223
LSI force ratio i/u				
Quadriceps 60 deg/s	90.8 ± 13.4	93.5 ± 12.6	91.1 ± 13.7	0.621
Quadriceps 180 deg/s	91.5 ± 16.2	94.6 ± 12.7	92.4 ± 16.5	0.618
Quadriceps 300 deg/s	90.7 ± 14.5	96.3 ± 19.6	90.7 ± 16.2	0.217
Hamstrings 60 deg/s	100.9 ± 20.0	97.3 ± 18.7	104.1 ± 15.3	0.437
Hamstrings 180 deg/s	102.6 ± 21.7	104.9 ± 13.7	110.0 ± 19.8	0.377
Hamstrings 300 deg/s	107.0 ± 24.8	105.3 ± 22.7	108.4 ± 21.0	0.897
Time from injury to surgery, days	138.1 ± 84.4	131.0 ± 75.3	110.1 ± 52.0	0.823
Operating time, min	63.1 ± 12.1	62.3 ± 17.1	47.7 ± 12.4	<0.001

TABLE 1 (Continued)

	QT ($n = 50$)	BPTB ($n = 42$)	HT ($n = 44$)	p-value
Accompanying injury noted preoperatively				
Partial medial meniscectomy	0 (0%)	4 (9%)	5 (10%)	0.057
Partial lateral meniscectomy	8 (19%)	7 (16%)	6 (12%)	0.918
Lateral femoral chondral lesion	2 (5%)	2 (5%)	1 (2%)	0.819
Medial femoral chondral lesion	0 (0%)	1 (2%)	4 (8%)	0.056
Lateral tibial chondral lesion	0 (0%)	2 (5%)	1 (2%)	0.301
Medial tibial chondral lesion	0 (0%)	2 (5%)	1 (2%)	0.301
Patellar chondral lesion	1 (2%)	2 (5%)	2 (4%)	0.730
Trochlear chondral lesion	0 (0%)	2 (5%)	0 (0%)	0.103

Note: Data are expressed as mean ± standard deviation or frequency (percentage).
Abbreviations: ACL, anterior cruciate ligament; ADL, activities of daily living; BPTB, bone-patellar tendon-bone; HT, hamstring tendon; IKDC, International Knee Documentation Committee; KOOS, Knee injury and Osteoarthritis Outcome Score; LSI, leg symmetry index; QoL, quality of life; QT, quadriceps tendon.

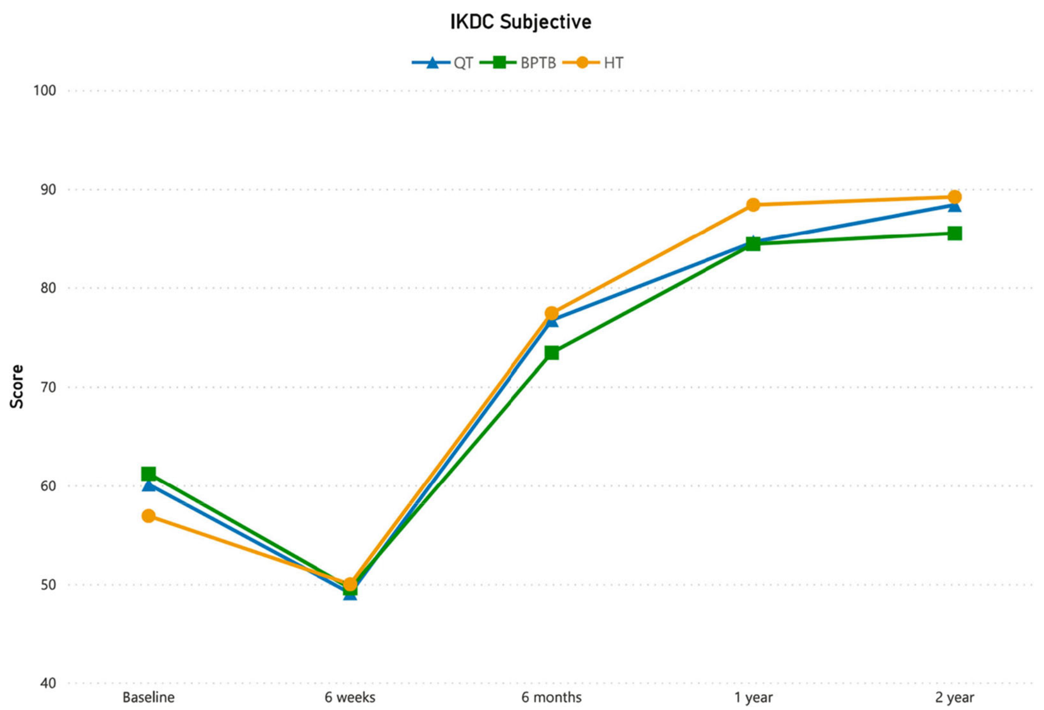

FIGURE 2 Mean outcome scores for International Knee Documentation Committee subjective at baseline and during follow-up for the different graft types. BPTB, bone-patellar tendon-bone; HT, hamstring tendon; QT, quadriceps tendon.

score for the BPTB-group compared to the HT-group ($p = 0.017$).

Primary outcome measure: IKDC subjective at 2-year follow-up

The lower boundary of the 95% CI for the QT-group (95% CI 85.4–91.5) fell within the predefined non-inferiority margin of –10 points, confirming non-inferiority of the QT autograft compared to both BPTB (95% CI 80.7–90.7) and HT (95% CI 85.0–93.8) autografts for IKDC subjective at 2-year follow-up (Figure A1, Table A1).

Furthermore, at 2-year follow-up, no statistically significant difference was found between groups for the mean IKDC subjective score: 88.4 ± 8.6 in the QT group, 85.5 ± 12.9 in the BPTB group, and 89.2 ± 12.1 in the HT ($p = 0.864$) (Figure 2, Table 2).

TABLE 2 Outcome measures: patient reported outcome over time.

	QT	BPTB	HT		*p*-value
IKDC subjective					
Baseline[a]	60.1 ± 13.7	61.2 ± 14.1	56.9 ± 13.1	Group	0.864
6 weeks[b]	49.1 ± 12.6	49.6 ± 11.7	50.0 ± 11.9	Time	<0.001
6 months[c]	76.7 ± 22.5	73.4 ± 10.1	77.4 ± 12.6	Time*Group	0.308
12 months[d]	84.6 ± 12.0	84.4 ± 13.2	88.4 ± 11.8		
24 months[e]	88.4 ± 8.6	85.5 ± 12.9	89.2 ± 12.1		
KOOS other symptoms					
Baseline[a]	68.2 ± 18.2	70.7 ± 23.5	67.1 ± 18.3	Group	0.670
6 weeks[f]	61.4 ± 14.8	64.7 ± 15.4	63.3 ± 16.9	Time	<0.001
6 months[g]	80.4 ± 13.0	83.5 ± 15.2	82.4 ± 16.2	Time*Group	0.933
12 months[h]	86.2 ± 10.5	87.3 ± 13.4	90.1 ± 11.4		
24 months[i]	89.5 ± 11.4	90.0 ± 12.8	90.4 ± 14.1		
KOOS pain					
Baseline[a]	73.4 ± 15.6	78.9 ± 12.6	74.0 ± 16.6	Group	0.838
6 weeks[j]	72.5 ± 15.3	73.4 ± 15.5	71.6 ± 13.9	Time	<0.001
6 months[k]	88.6 ± 10.2	87.5 ± 10.3	89.3 ± 11.1	Time*Group	0.311
12 months[h]	92.7 ± 9.2	92.3 ± 10.0	94.6 ± 8.5		
24 months[i]	92.7 ± 8.7	93.6 ± 9.6	93.6 ± 11.3		
KOOS ADL					
Baseline[a]	85.4 ± 14.3	90.3 ± 9.7	82.7 ± 14.6	Group	0.549
6 weeks[j]	75.9 ± 16.7	78.1 ± 14.6	79.8 ± 14.9	Time	<0.001
6 months[l]	93.1 ± 8.1	93.6 ± 7.6	95.0 ± 7.7	Time*Group	0.050
12 months[m]	96.7 ± 4.3	97.6 ± 5.4	98.2 ± 4.1		
24 months[i]	97.3 ± 4.1	97.4 ± 6.1	97.0 ± 8.9		
KOOS sport and recreation					
Baseline[a]	44.4 ± 23.0	46.1 ± 26.2	38.1 ± 27.0	Group	0.930
6 weeks[j]	30.5 ± 30.4	20.1 ± 22.7	24.6 ± 22.2	Time	<0.001
6 months[l]	66.0 ± 21.9	65.3 ± 21.9	68.4 ± 23.8	Time*Group	0.193
12 months [m]	80.4 ± 19.6	81.3 ± 17.1	86.5 ± 16.4		
24 months[i]	83.8 ± 17.4	81.2 ± 19.7	87.8 ± 16.2		
KOOS QoL					
Baseline[a]	47.3 ± 14.87	50.4 ± 18.1	46.7 ± 17.6	Group	0.366
6 weeks[j]	41.9 ± 15.8	44.3 ± 15.2	45.2 ± 14.5	Time	<0.001
6 months[l]	60.1 ± 16.6	61.5 ± 12.1	67.6 ± 16.9	Time*Group	0.116
12 months[m]	71.6 ± 16.0	71.1 ± 16.2	75.8 ± 19.0		
24 months[i]	80.8 ± 17.4	75.0 ± 17.5	83.3 ± 15.9		

TABLE 2 (Continued)

	QT	BPTB	HT		*p*-value
ACL-QOL					
Baseline[a]	45.4 ± 14.7	50.5 ± 17.4	49.1 ± 14.9	Group	0.073
6 weeks [f]	42.3 ± 15.4	45.5 ± 14.7	48.1 ± 15.3	Time	<0.001
6 months[n]	61.6 ± 19.8	66.1 ± 14.9	69.0 ± 16.2	Time*Group	0.866
12 months[d]	79.3 ± 19.4	84.0 ± 14.3	86.0 ± 14.8		
24 months[o]	82.8 ± 21.6	86.8 ± 14.7	88.2 ± 15.8		
Lysholm					
Baseline[a]	69.3 ± 14.7	71.3 ± 14.5	71.4 ± 14.7	Group	0.438
6 weeks[j]	63.3 ± 17.6	66.2 ± 13.8	66.3 ± 16.8	Time	<0.001
6 months[k]	83.5 ± 19.6	83.1 ± 11.0	86.4 ± 11.1	Time*Group	0.857
12 months[m]	87.1 ± 11.3	86.5 ± 10.5	89.4 ± 11.3		
24 months[i]	89.1 ± 11.4	88.0 ± 16.0	90.6 ± 14.0		
Tegner					
Baseline[a]	8.0 ± 1.2	7.9 ± 1.3	7.7 ± 1.1	Group	0.522
6 weeks[a]	2.9 ± 2.4	3.0 ± 2.5	3.9 ± 2.8	Time	<0.001
6 months[k]	8.6 ± 2.4	8.8 ± 1.2	8.9 ± 1.1	Time*Group	0.321
12 months[m]	7.4 ± 1.9	7.2 ± 2.3	7.4 ± 2.3		
24 months[i]	7.7 ± 2.2	7.5 ± 2.4	7.7 ± 2.6		
NRS satisfaction					
6 weeks[j]	8.1 ± 1.5	8.1 ± 2.5	8.5 ± 1.5	Group	0.695
6 months[p]	8.5 ± 2.6	7.9 ± 2.6	8.4 ± 1.9	Time	0.989
12 months[m]	8.7 ± 1.1	8.4 ± 2.6	8.8 ± 1.7	Time*Group	0.942
24 months[q]	8.7 ± 1.4	8.4 ± 1.5	9.1 ± 1.1		
6 weeks[j]	4.6 ± 2.0	4.7 ± 2.5	5.3 ± 1.9	Group	0.447
6 months[p]	6.2 ± 2.4	6.6 ± 2.2	6.8 ± 2.1	Time	<0.001
12 months[m]	7.2 ± 2.7	7.0 ± 2.8	7.7 ± 1.9	Time*Group	0.911
24 months[i]	7.7 ± 2.4	7.3 ± 3.0	7.8 ± 2.6		
VAS anterior knee pain at rest					
6 weeks[r]	1.2 ± 2.0	1.4 ± 2.1	1.0 ± 1.5	Group	0.426
6 months[s]	0.7 ± 0.9	0.7 ± 1.1	0.6 ± 1.1	Time	0.005
12 months[t]	0.5 ± 1.1	1.1 ± 2.6	0.3 ± 0.6	Time*Group	0.680
24 months[u]	0.2 ± 0.3	0.7 ± 1.1	0.7 ± 1.4		
VAS anterior knee pain exercise					
6 weeks[v]	3.7 ± 2.6	3.9 ± 2.6	3.6 ± 2.6	Group	0.302
6 months[w]	2.1 ± 2.1	3.1 ± 2.7	1.8 ± 1.9	Time	<0.001

(Continues)

TABLE 2 (Continued)

	QT	BPTB	HT		p-value
12 months[x]	1.3 ± 1.6	1.8 ± 2.9	0.8 ± 1.3	Time*Group	0.239
24 months[y]	0.8 ± 1.4	1.7 ± 2.0	1.4 ± 2.8		

Note: Data are expressed as mean ± standard deviation.

Abbreviations: ADL, activities of daily living; BPTB, bone-patellar tendon-bone; HT, hamstring tendon; IKDC, International Knee Documentation Committee; KOOS, Knee injury and Osteoarthritis Outcome Score; LSI, leg symmetry index; QoL, quality of life; QT, quadriceps tendon; VAS, visual analog scale.

[a]Analysis based on 50 patients with QT, 42 patients with PT and 44 patients with HT.
[b]Analysis based on 45 patients with QT, 35 patients with PT and 43 patients with HT.
[c]Analysis based on 45 patients with QT, 33 patients with PT and 39 patients with HT.
[d]Analysis based on 35 patients with QT, 30 patients with PT and 35 patients with HT.
[e]Analysis based on 33 patients with QT, 28 patients with PT and 31 patients with HT.
[f]Analysis based on 45 patients with QT, 34 patients with PT and 42 patients with HT.
[g]Analysis based on 44 patients with QT, 31 patients with PT and 37 patients with HT.
[h]Analysis based on 35 patients with QT, 30 patients with PT and 34 patients with HT.
[i]Analysis based on 30 patients with QT, 28 patients with PT and 31 patients with HT.
[j]Analysis based on 44 patients with QT, 34 patients with PT and 42 patients with HT.
[k]Analysis based on 43 patients with QT, 31 patients with PT and 37 patients with HT.
[l]Analysis based on 43 patients with QT, 30 patients with PT and 37 patients with HT.
[m]Analysis based on 35 patients with QT, 30 patients with PT and 33 patients with HT.
[n]Analysis based on 45 patients with QT, 31 patients with PT and 37 patients with HT.
[o]Analysis based on 32 patients with QT, 28 patients with PT and 31 patients with HT.
[p]Analysis based on 43 patients with QT, 30 patients with PT and 36 patients with HT.
[q]Analysis based on 30 patients with QT, 28 patients with PT and 30 patients with HT.
[r]Analysis based on 41 patients with QT, 32 patients with PT and 42 patients with HT.
[s]Analysis based on 38 patients with QT, 27 patients with PT and 34 patients with HT.
[t]Analysis based on: 30 patients with QT, 23 patients with PT and 31 patients with HT.
[u]Analysis based on 26 patients with QT, 19 patients with PT and 26 patients with HT.
[v]Analysis based on 43 patients with QT, 34 patients with PT and 42 patients with HT.
[w]Analysis based on 41 patients with QT, 29 patients with PT and 35 patients with HT.
[x]Analysis based on 31 patients with QT, 28 patients with PT and 31 patients with HT.
[y]Analysis based on 27 patients with QT, 23 patients with PT and 27 patients with HT.

Secondary outcome measures

PROMs

All PROMs improved significantly over time ($p < 0.001$), except NRS satisfaction scores which remained high (>8) and stable throughout follow-up ($p = 0.989$). No significant between-group or interaction effects were found for PROMs (all $p > 0.05$) (Table 2).

At 2-year follow-up, the proportion of patients achieving the PASS for IKDC subjective was 87.9% in the QT group, 78.6% in the BPTB group and 90.3% in the HT group. For KOOS Pain, PASS was achieved by 63.3%, 71.4% and 83.9%, respectively; for KOOS Symptoms by 96.7%, 96.4% and 96.8%; for KOOS ADL by 50.0%, 60.7% and 64.5%; for KOOS Sport by 73.3%, 67.9% and 93.6%; for KOOS Quality of Life by 90.0%, 82.1% and 96.8%; and for the Lysholm score by 80.0%, 75.0% and 77.4%. A between-group difference was observed only for KOOS Sport ($p = 0.04$).

Physical examination

No significant differences in instrumented Lachman test or IKDC physical examination grade were found between groups at any timepoint. AP laxity measured with the instrumented Lachman test increased over time: from 1.2 ± 1.0 mm at 6 weeks to 1.9 ± 1.8 mm 2 years postoperatively in the QT group; from 0.9 ± 1.0 mm to 1.2 ± 1.0 mm in the BPTB-group and from 1.1 ± 1.1 to 1.6 ± 1.2 mm in the HT-group. There was no significant group*time interaction effect (Table A1).

Strength and hop tests

At 6 months postoperatively, the HT-group demonstrated significantly higher LSI values for quadriceps strength at 180°/s and 300°/s compared to the QT-group (12.3% and 11.4% higher, respectively) and BPTB-group (14.5% and 10.7% higher, respectively) (all $p \le 0.035$). A significant group*time interaction effect was present for quadriceps strength ($p < 0.029$), with the HT-group showing a smaller

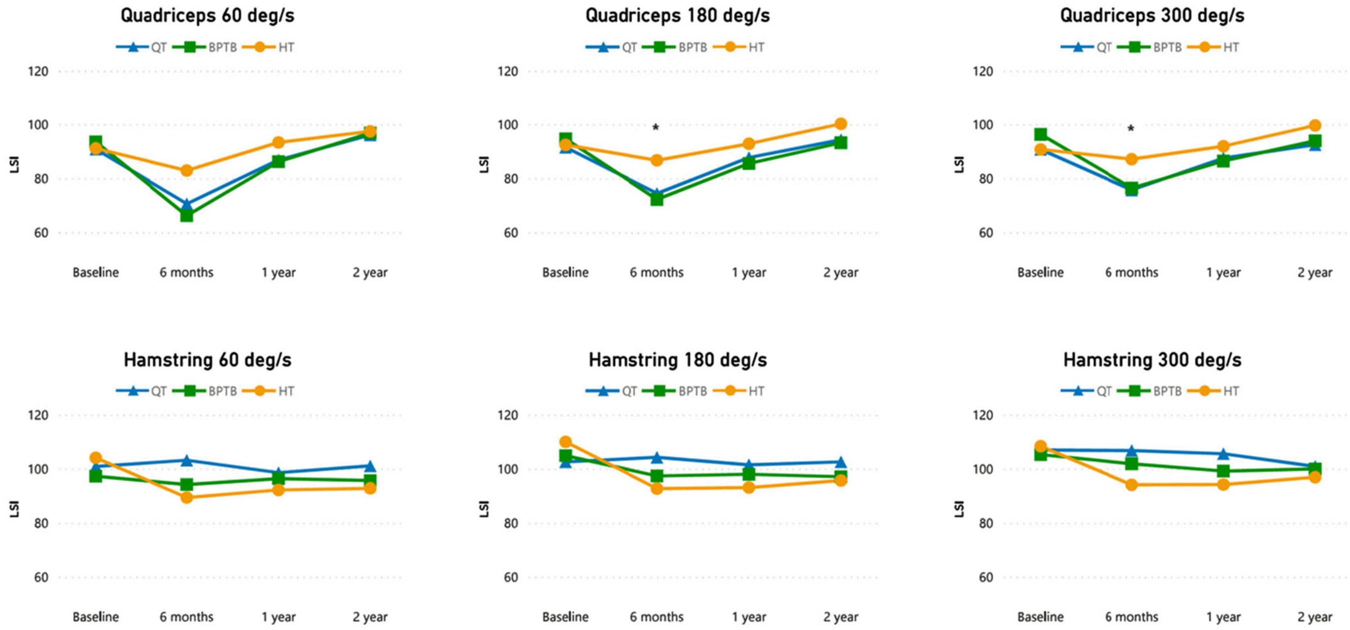

FIGURE 3 Mean outcome scores for limb symmetry index (LSI) for isokinetic strength at baseline and during follow-up for the different graft types. *Significant difference between HT versus QT and BPTB. BPTB, bone-patellar tendon-bone; HT, hamstring tendon; QT, quadriceps tendon.

initial decline and comparable values at 12 and 24 months (Figure 3 and Table A2). The HT-group also showed significantly higher LSI scores on the triple hop test at 6 months ($p = 0.011$), although this did not result in significant groups*time interaction. Hamstring strength at 60°/s and 300°/s did not improve significantly over time ($p = 0.120$ and 0.076, respectively). No other significant differences in LSI scores were observed across time or groups (Table A2).

Strength recovered to an LSI of ≥90% in 43.6%–85.7% for the QT-group, 29.7%–82.1% for the BPTB group and 55.2%–69% for the HT group (range depending on peak torque angles) at 1 year postoperative, and to an LSI of ≥90% in 53.1%–81.3%, 53.1%–75% and 51.4%–80%, respectively, at 2 years postoperative (Table A3).

Osteoarthritis

No significant between-group differences in radiographic osteoarthritis were observed ($p = 0.65$), and the majority of patients (≥87.1%) showed no osteoarthritis (KL grade 0) at 24 months (Table A1).

Adverse events

Four ipsilateral ACL clinical failures (8.0%) occurred in the QT-group, 5 (11.9%) in the BPTB-group and 2 (4.6%) in the HT-group (n.s.). Repeat surgeries other than for revision ACLR was needed in four patients from the QT-group (8.0%; two cyclops lesions, one case of residual synovitis with suspected bacterial infection but negative intraoperative cultures, one partial medial meniscectomy with hardware removal), in four patients from the BPTB-group (9.5%; two cyclops lesions, one partial meniscectomy, one hardware removal) and in three patients from the HT-group (6.8%; one hardware removal, one cyclops lesion with hardware removal, one microfracture trochlea) (n.s.). Additionally, one patient (2.0%) in the QT-group, three patients (7.1%) in the BPTB-group and five patients (11.4%) in the HT-group experienced symptoms of extension deficits, pain and swelling which either resolved spontaneously or did not require operative treatment (n.s.) (Table 3).

DISCUSSION

The most important finding of this study is that ACLR with an ipsilateral QT autograft was non-inferior to ACLR with an ipsilateral BPTB or HT autograft in terms of subjective patient-reported outcomes as measured with the IKDC subjective score at 2-year follow-up. These findings support the use of QT autografts as a viable alternative to BPTB and HT autografts in clinical practice [33].

Previous randomised trials comparing QT with either HT or BPTB have reported no significant differences in patient-reported outcomes [24, 34]. While multiple systematic reviews and meta-analyses have compared ACLR with a QT to ACLR with either a HT or BPTB [1, 7, 21, 22], only a few specifically used the IKDC subjective score to evaluate patient-reported outcomes. Our findings are consistent with these studies and extend them by directly comparing all three autografts in a single trial with

TABLE 3 Adverse events 2 years after anterior cruciate ligament reconstruction.

	QT (*n* = 50)	BPTB (*n* = 42)	HT (*n* = 44)	*p*-Value
Ipsilateral ACL rupture	4 (8.0%)	5 (11.9%)	2 (4.6%)	0.547
Contralateral ACL rupture	2 (4.0%)	1 (2.4%)	3 (6.8%)	0.596
Repeat surgery	4 (8.0%)	4 (9.5%)	3 (6.8%)	0.899
Abnormal symptoms: pain, swelling, extension deficits	1 (2.0%)	3 (7.1%)	5 (11.4%)	0.188
Other adverse events	2 (4.0%)	1 (2.4%)	1 (2.3%)	0.856

Note: Data are expressed as frequency (percentage).
Abbreviations: ACL, anterior cruciate ligament; BPTB, bone-patellar tendon-bone; HT, hamstring tendon; QT, quadriceps tendon.

uniform methodology. While only Kurkowski et al., in a meta-analysis including 27 studies with in total 659 patients treated with QT autograft, 2506 with BPTB autograft, and 2536 with HT autograft, reported significantly lower IKDC scores in the BPTB group compared to QT and HT, the absolute differences were small (mean 86.5 for QT, 84.4 for BPTB and 87.1 for HT) and not clinically relevant [6, 22]. Therefore, although their findings contrast with ours on a statistical level, the practical implications are limited. Their results were also based on longer-term follow-up with a minimum of 5 years. Meta-analyses by Dai et al. and Mouarbes et al. have shown comparable patient-reported outcomes across graft types that are consistent with those observed in our study [7, 26]. Together, these findings highlight that when directly compared using consistent methodology, patient-reported outcomes after ACLR appear to be similar across all commonly used autograft types.

The question of differential failure rates among HT, BPTB and QT autografts remains clinically relevant, despite similar patient-reported outcome. Recent registry data by Zhou et al. and several meta-analyses have reported comparable revision rates among graft types [7, 26, 50, 52]. In contrast, large Scandinavian registries and the MOON cohort demonstrated higher revision and failure rates, respectively, for HT compared with BPTB grafts, particularly in young, high-demand athletes [41, 32, 51]. Danish data further indicated an increased revision risk for ACLR with QT grafts (4.7%) relative to BPTB (1.5%) and HT (2.3%) grafts, again most pronounced in younger patients [23]. Collectively, these findings suggest that BPTB grafts may offer the lowest failure risk in young, active individuals. However, because BPTB autografts are contraindicated in skeletally immature patients, soft tissue autografts are preferentially used in the youngest group – who is at highest risk for failure [49]. Registry findings may therefore partly reflect a selection bias disadvantaging HT and QT grafts in the youngest and high-risk cohorts. Moreover, most registry-based data derive from ACL reconstructions performed without concomitant ALCR [45]. In contemporary practice, historical observations of graft failure risk, supported by high-level randomised evidence, have led to a consensus to perform ALCR in addition to ACLR [40]. Consequently, historical registry-based failure rates may not fully reflect current surgical practice, as ALCR may lower absolute graft failure rates across all ACL graft types and complicate direct comparisons between graft types [8].

Although in the present study a numerical difference in failure rates was observed across ACL graft types, this difference did not reach statistical significance and, in the context of premature study termination and limited statistical power, reflects imprecision rather than evidence of a true effect. Therefore, the observed findings are inconclusive and do not allow firm clinical conclusions regarding graft-specific differences in failure rates.

This study observed a significant group-by-time interaction for quadriceps strength, with the HT group showing significantly higher LSI values at 180 and 300°/s at 6 months compared to both QT and BPTB. Additionally, the HT group achieved significantly higher LSI scores on the triple hop test at 6 months. These differences likely reflect early-phase donor site morbidity associated with harvesting grafts from the extensor mechanism, as previously reported in the literature [17, 19]. Although these differences were statistically significant at 6 months, they diminished over time, and all groups showed comparable strength at 1- and 2-year follow-up. Our findings align with previous studies showing that HT autografts may better preserve early quadriceps function, while BPTB and QT may induce early extensor strength deficits that typically normalise over time [17, 19]. Nonetheless, full quadriceps strength recovery to a LSI of 90% was not consistently achieved by 2 years in any group, with approximately only half of patients meeting an LSI ≥ 90% in either group, consistent with previous studies [19, 48]. Therefore, restoration of quadriceps strength should remain a key focus of rehabilitation, particularly in patients treated with QT or BPTB grafts [14]. Clinicians may consider this when counselling patients with performance goals within the first postoperative year.

Patient-reported instability, measured by VAS, improved significantly over time, with no significant differences between graft types or interaction effects. This supports the interpretation that all grafts restored functional stability to a comparable degree. Satisfaction with surgical outcome, as measured by NRS, was high across all groups at all time points, without relevant differences or time effects. These findings further confirm that graft choice did not influence patients' subjective perception of success.

Donor site morbidity, particularly anterior knee pain, is a longstanding concern with BPTB grafts [1, 35]. In this study, however, anterior knee pain levels were low, and did not significantly differ between groups at any time point. This likely reflects the effect of modern, minimally invasive harvesting techniques used in all three participating centres. Our findings support those of previous studies suggesting that anterior knee pain might no longer be considered a primary disadvantage of QT or BPTB grafts when harvested carefully [3, 12]. With optimised techniques, donor site morbidity may no longer be a decisive factor in graft selection. Surgeons should weigh graft-specific differences in donor site morbidity against other functional outcomes and patient priorities.

Current evidence suggests that use of a QT autograft for ACL reconstruction may be associated with an increased risk of osteoarthritis compared with HT autografts; [30] however, available long-term data are limited to define the magnitude or mechanisms of this risk [7, 22, 47, 52]. As the present manuscript reports predefined 2-year outcomes and no differences were observed at this time point, no conclusions regarding long-term degenerative changes can be drawn.

This study has several limitations. First, the study was underpowered for the originally defined primary outcome of graft failure. Second, the study was conducted in a limited number of high-volume centres, which may limit generalisability. Third, patients and assessors were not blinded, although the use of validated, PROMs reduces the risk of bias. Fourth, graft-specific ACL reconstruction techniques, including drilling approach, fixation method and fixation angle, were used; although this may be considered a limitation, these techniques reflect contemporary clinical practice and therefore enhance the external validity of the study. Finally, additional procedures such as anterolateral corner reconstruction were not permitted, which may affect applicability in cases with high-grade pivot shift. Despite these limitations, this study contributes to the existing literature by providing a prospective randomised comparison of three autograft types within a single study design. Future trials should compare these grafts in larger, multicentre designs and consider the role of combined procedures such as ALCR, particularly for patients with rotational instability.

CONCLUSION

The use of QT autograft is non-inferior to BPTB or HT autograft in isolated ACLR in terms of patient-reported outcomes as measured with the IKDC subjective score 2 years postoperatively. The findings of the present study support the use of the QT autograft as a valid alternative to BPTB and HT autograft for ACLR and may assist surgeons in individualising graft choice based on patient characteristics, preferences, and rehabilitation needs.

AUTHOR CONTRIBUTIONS

All the authors have been actively involved in the planning and enactment of the study and are involved in writing of the submitted paper.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS STATEMENT

Medical Ethical Committee (MEC-U); NL52749.044.16. Written informed consent was obtained for all patients prior to enrollment.

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APPENDIX

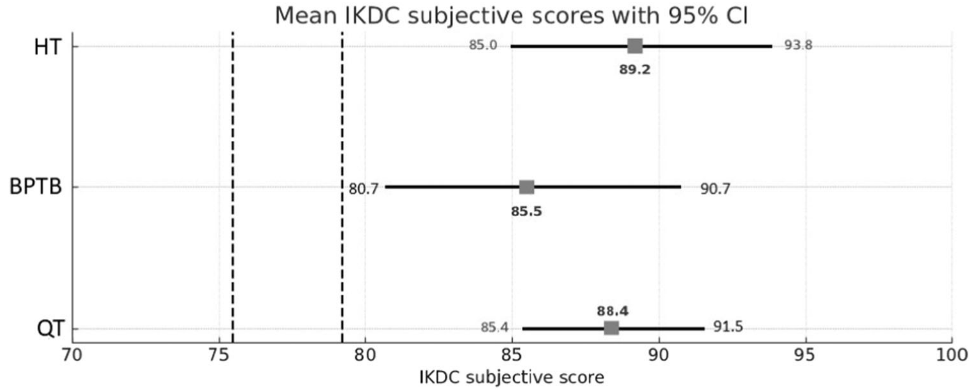

FIGURE A1 Noninferiority International Knee Documentation Committee (IKDC) subjective score results at 2-year follow-up. Data are expressed as mean with 95% confidence interval. Dotted lines indicate the non-inferiority margin of the IKDC subjective score of HT and BPBT group, the clinically relevant difference is –10 points. BPTB, bone-patellar tendon-bone; HT, hamstring tendon; QT, quadriceps tendon.

TABLE A1 Outcome measures: Clinimetry and radiography over time.

		QT	BPTB	HT		*p*-value
IKDC physical examination						
Baseline	A	0 (0%)	1 (2.4%)	0 (0%)		0.612
	B	4 (8%)	2 (4.8%)	4 (9.1%)		
	C	42 (84%)	32 (76.2%)	34 (77.3%)		
	D	4 (8%)	7 (16.7%)	6 (13.6%)		
6 weeks	A	2 (4.3%)	5 (12.8%)	5 (11.9%)		0.693
	B	27 (57.4%)	16 (41.0%)	22 (52.4%)		
	C	9 (19.1%)	10 (25.6%)	8 (19%)		
	D	9 (19.1%)	8 (20.5%)	7 (16.7%)		
6 months	A	15 (32.6%)	13 (31.7%)	15 (35.7%)		0.294
	B	16 (34.8%)	13 (31.7%)	21 (50%)		
	C	13 (28.3%)	12 (29.3%)	5 (11.9%)		
	D	2 (4.3%)	3 (7.3%)	1 (2.3%)		
12 months	A	19 (48.7%)	21 (55.3%)	23 (56.1%)		0.864
	B	14 (35.9%)	12 (31.6%)	12 (29.3%)		
	C	5 (12.8%)	5 (13.2%)	6 (16.6%)		
	D	1 (2.6%)	0 (0%)	0 (0%)		
24 months	A	12 (36.4%)	17 (51.5%)	18 (52.9%)		0.409
	B	12 (36.4%)	7 (21.2%)	11 (32.4%)		
	C	9 (27.3%)	9 (27.3%)	5 (14.7%)		
	D	0 (0%)	0 (0%)	0 (0%)		
Lachman, delta						
6 weeks[a]		1.2 ± 1.0	0.9 ± 1.0	1.1 ± 1.1	Group	0.355
6 months[b]		1.7 ± 1.2	1.4 ± 1.0	1.3 ± 1.3	Time	<0.001
12 months[c]		1.9 ± 1.4	1.4 ± 1.2	1.5 ± 1.3	Time*Group	0.147
24 months[d]		1.9 ± 1.8	1.2 ± 1.0	1.6 ± 1.2		
Rontgen KL score						
Baseline	*0*	48 (96%)	40 (97.6%)	43 (97.7%)		0.861
	1	2 (4%)	1 (2.4%)	1 (2.3%)		
12 months	*0*	36 (94.7%)	29(90.6%)	34 (94.4%)		0.739
	1	2 (5.3%)	3 (9.4%)	2 (5.6%)		
24 months	*0*	31 (96.9%)	29 (84.4%)	23 (85.2%)		0.219
	1	1 (3.1%)	5 (15.2%)	4 (14.8%)		

Note: Data are expressed as mean ± standard deviation or frequency (percentage).
Abbreviations: BPTB, bone-patellar tendon-bone; HT, hamstring tendon; IKDC, International Knee Documentation Committee; KL, Kellgren and Lawrence; QT, quadriceps tendon.
[a]Analysis based on 36 patients with QT, 34 patients with PT and 39 patients with HT.
[b]Analysis based on 46 patients with QT, 39 patients with PT and 42 patients with HT.
[c]Analysis based on 39 patients with QT, 37 patients with PT and 40 patients with HT.
[d]Analysis based on 33 patients with QT, 32 patients with PT and 35 patients with HT.

TABLE A2 Outcome measures: LSI for isokinetic strength and hop tests over time.

	QT	BPTB	HT		*p*-value
Quadriceps 60 deg/s					
Baseline[a]	90.8 ± 13.4	93.5 ± 12.6	91.1 ± 13.7	Group	0.061
6 months[b]	70.6 ± 19.0	66.2 ± 18.3	82.9 ± 18.2	Time	<0.001
12 months[c]	86.8 ± 14.8	86.2 ± 16.1	93.3 ± 17.4	Time*Group	0.005
24 months[d]	96.1 ± 16.0	96.8 ± 12.5	97.4 ± 13.5		
Quadriceps 180 deg/s					
Baseline[a]	91.5 ± 16.2	94.6 ± 12.7	92.4 ± 16.5	Group	0.005
6 months[b]	74.4 ± 18.4	72.2 ± 17.3	86.7 ± 16.4	Time	<0.001
12 months[c]	87.7 ± 12.9	85.5 ± 15.1	92.8 ± 14.7	Time*Group	0.029
24 months[d]	94.3 ± 11.1	93.2 ± 10.1	100.2 ± 11.2		
Quadriceps 300 deg/s					
Baseline[a]	90.7 ± 14.5	96.3 ± 19.6	90.7 ± 16.2	Group	0.035
6 months[b]	75.7 ± 17.1	76.4 ± 20.8	87.1 ± 17.0	Time	<0.001
12 months[c]	87.5 ± 12.8	86.4 ± 17.4	91.9 ± 12.9	Time*Group	0.014
24 months[d]	92.5 ± 12.7	93.9 ± 11.9	99.6 ± 14.2		
Hamstring 60 deg/s					
Baseline[e]	100.9 ± 20.0	97.3 ± 18.7	104.1 ± 15.3	Group	0.065
6 months[f]	103.2 ± 14.1	94.2 ± 16.0	89.4 ± 13.7	Time	0.120
12 months[g]	98.6 ± 11.1	96.4 ± 12.3	92.2 ± 12.8	Time*Group	0.117
24 months[d]	101.1 ± 17.2	95.7 ± 14.9	92.8 ± 16.1		
Hamstring 180 deg/s					
Baseline[e]	102.6 ± 21.7	104.9 ± 13.7	110.0 ± 19.8	Group	0.310
6 months[f]	104.3 ± 17.2	97.4 ± 13.2	92.7 ± 19.1	Time	0.009
12 months[g]	101.5 ± 13.6	98.0 ± 13.2	93.1 ± 14.2	Time*Group	0.064
24 months[d]	102.6 ± 16.5	97.1 ± 14.2	95.7 ± 19.2		
Hamstring 300 deg/s					
Baseline[e]	107.0 ± 24.8	105.3 ± 22.7	108.4 ± 21.0	Group	0.359
6 months[f]	106.8 ± 19.6	101.9 ± 25.1	94.1 ± 19.5	Time	0.076
12 months[g]	105.6 ± 19.1	99.2 ± 15.7	94.2 ± 24.0	Time*Group	0.366
24 months[d]	100.9 ± 19.73	100.0 ± 17.2	96.89 ± 25.2		
Single leg hop					
6 months[h]	80.8 ± 31.5	78.8 ± 24.8	91.7 ± 18.8	Group	0.116
12 months[i]	96.1 ± 11.9	98.6 ± 8.8	99.1 ± 17.9	Time	<0.001
24 months[j]	97.6 ± 8.4	98.9 ± 6.1	99.4 ± 5.8	Time*Group	0.277
Triple leg hop					
6 months[k]	82.4 ± 30.8	81.5 ± 23.9	93.9 ± 17.4	Group	0.011
12 months[l]	97.3 ± 9.5	96.0 ± 7.0	99.6 ± 5.1	Time	<0.001
24 months[m]	97.6 ± 4.7	98.4 ± 4.6	100.3 ± 6.8	Time*Group	0.295

TABLE A2 (Continued)

	QT	BPTB	HT		*p*-value
Side hop					
6 months[n]	80.0 ± 35.6	78.4 ± 29.1	93.5 ± 20.6	Group	0.193
12 months[o]	95.1 ± 15.7	98.3 ± 17.7	98.4 ± 9.6	Time	<0.001
24 months[p]	97.7 ± 24.7	98.4 ± 14.5	97.5 ± 11.5	Time*Group	0.169

Note: Data are expressed as mean ± standard deviation.
Abbreviations: BPTB, bone-patellar tendon-bone; HT, hamstring tendon; LSI, leg symmetry index; QT, quadriceps tendon.
[a]Analysis based on 50 patients with QT, 42 patients with PT and 44 patients with HT.
[b]Analysis based on 46 patients with QT, 39 patients with PT and 44 patients with HT.
[c]Analysis based on 39 patients with QT, 37 patients with PT and 40 patients with HT.
[d]Analysis based on 32 patients with QT, 32 patients with PT and 35 patients with HT.
[e]Analysis based on 30 patients with QT, 26 patients with PT and 22 patients with HT.
[f]Analysis based on 30 patients with QT, 28 patients with PT and 22 patients with HT.
[g]Analysis based on 28 patients with QT, 28 patients with PT and 29 patients with HT.
[h]Analysis based on 38 patients with QT, 31 patients with PT and 36 patients with HT.
[i]Analysis based on 39 patients with QT, 35 patients with PT and 38 patients with HT.
[j]Analysis based on 31 patients with QT, 31 patients with PT and 31 patients with HT.
[k]Analysis based on 36 patients with QT, 30 patients with PT and 34 patients with HT.
[l]Analysis based on 39 patients with QT, 35 patients with PT and 37 patients with HT.
[m]Analysis based on 31 patients with QT, 31 patients with PT and 30 patients with HT.
[n]Analysis based on 35 patients with QT, 28 patients with PT and 33 patients with HT.
[o]Analysis based on 39 patients with QT, 34 patients with PT and 36 patients with HT.
[p]Analysis based on 30 patients with QT, 30 patients with PT and 30 patients with HT.

TABLE A3 Outcome measures: Strength recovery to a limb symmetry index of ≥90%.

	QT	BPTB	HT	*p*-value
1 year postoperative				
Quadriceps 60 deg/s	17 (43.6%)	16 (43.2%)	25 (62.5%)	0.148
Quadriceps 180 deg/s	21 (53.8%)	11 (29.7%)	23 (57.5%)	0.031
Quadriceps 300 deg/s	18 (46.2%)	14 (37.8%)	24 (60%)	0.143
Hamstring 60 deg/s	22 (78.6%)	19 (67.9%)	16 (55.2%)	0.170
Hamstring 180 deg/s	21 (75%)	20 (71.4%)	17 (58.6%)	0.375
Hamstring 300 deg/s	24 (85.7%)	23 (82.1%)	20 (69%)	0.263
2 years postoperative				
Quadriceps 60 deg/s	22 (68.8%)	24 (75%)	25 (71.4%)	0.856
Quadriceps 180 deg/s	19 (59.4%)	17 (53.1%)	28 (80%)	0.053
Quadriceps 300 deg/s	17 (53.1%)	17 (53.1%)	26 (74.3%)	0.120
Hamstring 60 deg/s	26 (81.3%)	21 (65.6%)	18 (51.4%)	0.037
Hamstring 180 deg/s	26 (81.3%)	22 (68.8%)	21 (60%)	0.166
Hamstring 300 deg/s	22 (68.8%)	23 (71.9%)	22 (62.9%)	0.724

Note: Data are expressed as frequency (percentage).
Abbreviations: BPTB, bone-patellar tendon-bone; HT, hamstring tendon; QT, quadriceps tendon.